

RCME

Installation and operating instructions



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**Warning**

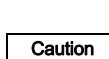
Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document**Warning**

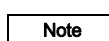
If these safety instructions are not observed, it may result in personal injury.

**Warning**

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.

**Caution**

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

**Note**

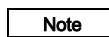
Notes or instructions that make the job easier and ensure safe operation.

2. Scope of these instructions

These installation and operating instructions apply to the Grundfos RCME rainwater pumping system. For instructions not mentioned specifically in this manual, see the installation and operating instructions supplied with the SB feed pump.

3. Delivery and handling

The Grundfos RCME pressure-boosting unit, including all sub-components, is delivered from factory on a wooden pallet designed for transport by fork-lift truck or a similar vehicle.

**Note**

To ensure safe transport, we recommend that the system is transported with suitable lifting tools.

4. Applications

The Grundfos RCME rainwater system is designed to supply collected rainwater to toilets and laundries in large buildings such as:

- shopping malls
- schools
- office buildings
- hospitals and nursing homes
- greenhouses and farms
- multi-family homes, housing complexes, etc.

5. Product description

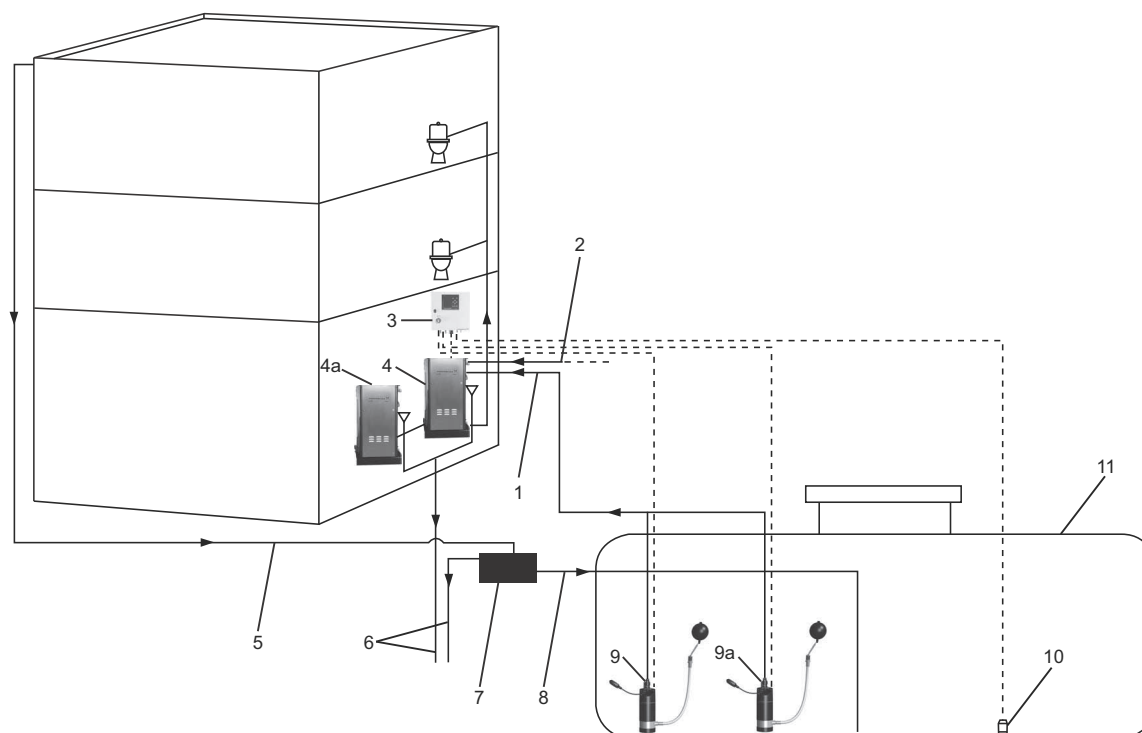


Fig. 1 Installation example

Pos.	Description
1	Rainwater supply to buffer tank(s)
2	Mains water supply to buffer tank(s)
3	Control cabinet with CU 372 control unit
4	RCME pressure-boosting unit
4a	Additional buffer tank (option)
5	Rainwater supply pipe
6	Overflow to sewer
7	Rainwater filter unit
8	Rainwater supply to collecting tank
9	SB feed pump with floating suction strainer
9a	Additional SB feed pump (option)
10	Hydrostatic pressure sensor
11	Rainwater collecting tank

The Grundfos RCME rainwater system consists of a pressure-boosting unit and a control cabinet with a Grundfos CU 372 control unit. The system also includes a Grundfos SB submersible pump which functions as feed pump for the RCME pressure-boosting unit. Install the feed pump in a rainwater collecting tank.

The RCME unit incorporates back-up of mains water supply. This function is automatically activated if the collecting tank is emptied, for example during long periods without rain.

The RCME system is operated and monitored via the user-friendly control panel on the CU 372 control unit that monitors and controls the rainwater system.

The RCME system can communicate in many different ways, for example with a building management system.

5.1 Overview of standard system components

The RCME system consists of the following main components: Position numbers refer to fig. 2, page 25.

- RCME 5-4 pressure-boosting unit
 - CME 5-4 booster pump (pos. 19)
 - non-return valve (pos. 9)
 - buffer tank (pos. 22) with level sensor (pos. 5)
 - integrated pressure vessel (pos. 21)
 - mains water valve (pos. 12)
 - steel base plate (pos. 18) with isolating valve (pos. 14)
- control cabinet with control unit (pos. 24)
- SB 3-45 submersible feed pump with floating suction strainer (pos. 20)
- hydrostatic pressure sensor for collecting tank (pos. 8)
- flow sensor (pos. 7)
- discharge hose (pos. 15).

5.1.1 Options

The following options are available for the system:

- additional buffer tank
- additional SB 3-45 feed pump.

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6. Identification

6.1 Nameplate

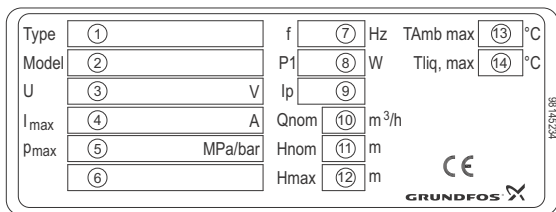


Fig. 2 Nameplate

Pos.	Description
1	Type designation
2	Model
3	Supply voltage [V]
4	Maximum current [A]
5	Maximum pressure [MPa/bar]
6	Country of origin
7	Frequency [Hz]
8	Input power [W]
9	Enclosure class
10	Rated flow [m³/h]
11	Head at rated flow [m]
12	Maximum head [m]
13	Maximum ambient temperature [°C]
14	Maximum liquid temperature [°C]

6.2 Type key

Example	R	CME	05	03
Rainwater				
Pump type CME: Centrifugal Modular booster pump with integrated frequency converter				
Rated flow rate Rated flow rate [m³/h]				
Number of phases of the electrical supply				

7. Mechanical installation



Warning

Installation and operation must comply with local regulations and accepted codes of good practice.

Checks before installation:

- Does the product correspond to order?
- Are all visible parts intact?

7.1 Location

7.1.1 RCME unit

The RCME unit should be sited in a dry, well-ventilated and frost-free position where it will not be exposed to extremes of temperature. In order to reduce bacteria growth, the ambient temperature should ideally be less than 20 °C. The unit may be located outdoors in a weather-, frost- and rodent-proof enclosure with adequate ventilation, especially during hot weather.

The RCME unit must not be installed into roof spaces.

The vibration level associated with any rotating equipment will cause disturbance, and if a leak occurs, considerable water damage could result.

If the RCME unit is installed in a basement and the buffer tank is installed below the collecting tank, siphoning may occur from the collecting tank to the buffer tank. In order to prevent this, an anti-siphoning valve should be installed or the pipework should be designed in such a way that siphoning is prevented.

Provide drainage facilities for the unit for use in case of leakage, overflow or loss of water during commissioning and service.

Position the RCME unit to allow access for maintenance and service.

Provide adequate drainage facilities and protection from water damage in the immediate vicinity of the RCME unit.

7.1.2 Control cabinet

The control cabinet is designed for wall mounting.

The CU 372 must not be exposed to direct sunlight.

7.1.3 SB 3-45 feed pump

Install the feed pump according to the installation and operating instructions supplied with the pump.

7.1.4 Hydrostatic pressure sensor for collecting tank

The sensor cable must not be kinked or pinched as this can result in faulty measurements.

Caution **Max. bending radius: 120 mm.**

Ensure that moisture does not enter the sensor cable sheath.

Place the hydrostatic pressure sensor close to the bottom of the collecting tank.

7.2 Foundation

Position the RCME unit on the supplied base plate. The base plate must be placed on an even and solid surface capable of supporting the weight of the unit with filled buffer tank, for instance on a concrete floor or foundation.

See section [19.7 Weight of RCME unit](#).

7.3 Pipework

Caution All external pipes must be properly supported before they are connected to the RCME unit to ensure that the internal pipework is not stressed.

A non-return valve is fitted on the booster pump discharge manifold, see pos. 9 on fig. 2, page 25.

Note All pipework subject to freezing conditions must be adequately insulated and heat-traced if necessary.

7.3.1 Rainwater inlet pipe from collecting tank

For thread size and location, see fig. 1, page 24, and fig. 3, page 25.

7.3.2 Mains water supply pipe and flow sensor

Position numbers refer to fig. 2, page 25.

Mount the flow sensor (pos. 7) in the mains water inlet pipe according to the guidelines described in the instructions delivered with the flow sensor.

7.3.3 Discharge pipe

Position numbers refer to fig. 2, page 25.

Connect the discharge hose (pos. 15) between the booster pump discharge manifold and the isolating valve (pos. 14).

Connect piping from the building to the isolating valve. For thread size see fig. 3, page 25.

7.3.4 Overflow

Lead the overflow to a sewer via a funnel to ensure that there is an air gap between the buffer tank and the pipe leading to the sewer. See fig. 5.



Fig. 3 Example of overflow arrangement

7.4 Additional feed pump

Install an additional feed pump so that it pumps rainwater into the same supply pipe as the other feed pump. See Fig. 1, page 3.

7.5 Additional buffer tank

Install an additional buffer tank by connecting it to the main buffer tank. See fig. 4.

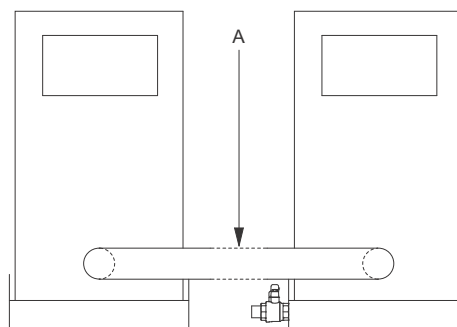


Fig. 4 Connection pipe (A) installed between main buffer tank and additional buffer tank

8. Electrical installation

Warning

The electrical connection must be carried out in accordance with local regulations.



Before making any connections to the unit or the feed pump, switch off the power supply, and make sure that it cannot be accidentally switched on.

We recommend fitting an earth leakage circuit breaker (ELCB) to the incoming power supply.

Check that the RCME unit and feed pump are suitable for the power supply at the installation site.

All electrical connections must comply with enclosure class IP54.

Caution

8.1 Electrical connection

The following connections must be made before the system can be commissioned.

8.1.1 Connection of power supply

Check that the conductor cross-section of the power supply cable used is suitable for the maximum current (I_{max}) stated on the nameplate of the RCME unit.

Caution

Connect the power supply as shown in the wiring diagram supplied with the control cabinet.

8.1.2 Connection of hydrostatic pressure sensor for collecting tank

Connect the supplied hydrostatic pressure sensor as shown in the wiring diagram supplied with the control cabinet.

Enter the tank depth and desired low-level limit for the collecting tank in display 4.1. See section 15.1 Tank settings (4.1).

8.1.3 Connection of feed pump

Connect the feed pump as shown in the wiring diagram supplied with the control cabinet.

8.1.4 Connection of flow sensor

Connect the flow sensor as shown in the wiring diagram supplied with the control cabinet.

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8.2 Motor protection

8.2.1 CME 5-4 booster pump

The motor incorporates thermal protection against slow overloading, blocking and dry running.

8.2.2 SB 3-45 feed pump

The feed pump is protected by a motor-protective circuit breaker with automatic resetting. The motor-protective circuit breaker is located inside the control cabinet and has been set from factory.

8.3 EMC-correct installation

In order to ensure an EMC-correct installation, these guidelines for communication and signal cables must be followed:

- Leave the screen of the cable as close to the connecting terminals as possible.
- Fix the screen with a cable clamp.

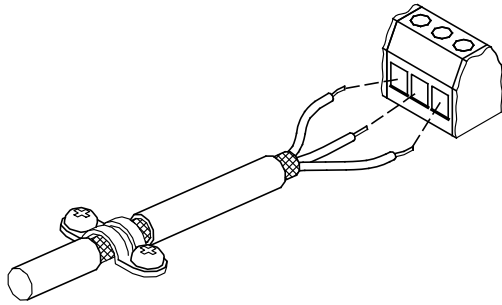


Fig. 5 Example of stripped cable with screen

- Connect the screen of communication and signal cables to frame at both ends.
- Screws for frame connections must always be tightened whether a cable is fitted or not.

9. Control panel

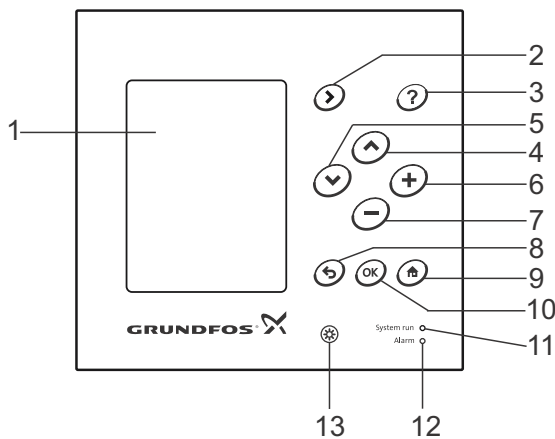


Fig. 6 CU 372 control panel

Pos.	Button/ indicator light	Description
1		LCD display
2		Go to next column in menu structure Press [→] to go to the next column in the menu structure. If [→] is pressed in the last column, it changes to the first column to the left.
3		Go to help text Press [?] to display the relevant help text. All the elements in the display have a selectable help text. Press [?] again to close the help text window.
4		Move up in lists Press [↑] to move up in lists. Press [↑] in the top line of the list to move to the bottom of the list.
5		Move down in lists Press [↓] to move down in lists. Press [↓] in the bottom line of the list to move to the top of the list.
6		Increase the value of a selected parameter Press [+] to increase the value of a selected parameter.
7		Decrease the value of a selected parameter Press [-] to decrease the value of a selected parameter.
8		Go one display back Press the button to return to the previous display in the menu. Press this button to cancel a change.
9		Return to the "Status" menu Press the button to return to the system overview display.
10		Save a value Use [OK] as Enter button. Also use [OK] to begin the process of changing a value and to save the change.
11		Green indicator light (operation) The green indicator light is on when the power supply has been switched on.
12		Red indicator light (alarm) The red indicator light is on if there is an alarm or a warning. The fault can be identified from the alarm list.
13		Change the contrast of the display Press the contrast button to change the display contrast. Press [+] or [-] to increase or decrease the contrast.

Active buttons are backlit.

9.1 Display layout

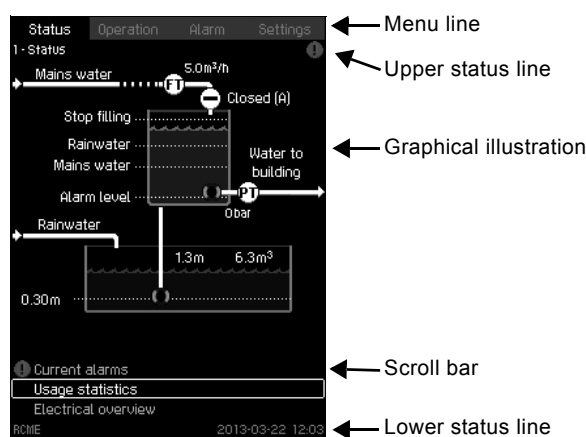


Fig. 7 Display layout

Menu line

The display has four main menus:

Status	Indication of system status
Operation	Changes to operating parameters (password option)
Alarm	Alarm log for fault-finding purposes
Settings	Changes to system settings (password option)

Upper status line

The upper status line shows the following:

- the display number and name (left side)
- the symbol ⓘ in case of alarm (right side)
- the symbol ⚠ in case of warning (right side)
- the symbol ⚙ if the service language has been selected (right side).

Lower status line

The lower status line shows the following:

- system name (left side)
- date and time (right side).

Graphical illustration

The graphical illustration shows the rainwater system, including feed pump and collecting tank. The display also shows various settings as well as system status, a historical indication or other elements.

Scroll bar

If the list of elements exceeds the display, the symbols ▲ and ▼ will appear in the scroll bar to the right. Use [▲] and [▼] to move up and down in the list.

10. Commissioning

10.1 Checks before first start-up

Check the following before the system is started up for the first time:

- Mechanical installation has been made as described in section [7. Mechanical installation](#).
- Electrical installation has been made as described in section [8. Electrical installation](#).
- All isolating valves are open.
- The precharge pressure of the integrated pressure vessel is set to 70 % of the desired setpoint.
For further information, see section [17.3 Integrated pressure vessel](#).
- The collecting tank has been filled sufficiently with water in order for the SB feed pump to deliver water to the buffer tank.

10.2 Startup

Position numbers refer to fig. 2, page 25.

1. Switch on the power supply, and wait for the first display to appear.
 2. Set "Display language", "Units", "Date and time" in displays 4.7.1 to 4.7.3.
See sections [15.7.1 Display language \(4.7.1\)](#) to [15.7.3 Date and time \(4.7.3\)](#).
 3. Make the collecting tank settings.
See section [15.1 Tank settings \(4.1\)](#).
 4. Check the maximum permissible discharge pressure for the booster pump (p_{max}) and adjust, if necessary.
See section [15.2 Booster pump settings \(4.2\)](#).
 5. Clear any alarms.
See section [14.1 Current alarms \(3.1\)](#).
 6. Select and define the conditions to be monitored.
See section [15.3.1 Digital inputs \(4.3.1\)](#).
 7. Set the feed pump to "Auto".
See section [13. Operation \(2\)](#).
 8. Set the mains water valve to "Auto".
See section [13. Operation \(2\)](#).
 9. Check that the feed pump stops and the mains water valve closes when the buffer tank has been filled to the stop level.
 10. Prime the booster pump:
 11. Close the isolating valve on the base plate outlet (pos. 14).
 12. Remove the front cover (pos. 23) of the RCME unit.
 13. Place a receptacle beneath the drain valve (pos. 26) mounted on the booster pump discharge manifold to receive drain water.
 14. Open the drain valve (pos. 26).
 15. Set the booster pump to "On".
See section [13. Operation \(2\)](#).
 16. Close the drain valve (pos. 26) when a steady stream of water comes out of the drain.
 17. The booster unit is now primed.
 18. Open the isolating valve (pos. 14) on the base plate outlet.
 19. Set the booster pump to "Auto".
See section [13. Operation \(2\)](#).
 20. Activate rainwater consuming equipment, for example toilets or garden hose, in order to prime the system and pipework.
- The system is now ready to supply rainwater to the building.

11. Menus

Functions tree

1. Status (page 9)

- 1.1 Electrical overview (page 9)
 - 1.1.1 Analog inputs (page 9)
 - 1.1.2 Digital inputs (page 9)
 - 1.1.3 Digital outputs (page 10)
- 1.2 Usage statistics (page 10)

2. Operation (page 10)

|

3. Alarm (page 10)

- 3.1 Current alarms (page 11)
- 3.2 Alarm log (page 11)

4. Settings (page 12)

- 4.1 Tank settings (page 12)
- 4.2 Booster pump settings (page 13)
- 4.3 Inputs and outputs (page 13)
 - 4.3.1 Digital inputs (page 14)
 - 4.3.2 Analog inputs (page 15)
 - 4.3.3 Digital outputs (page 15)
 - 4.3.4 Counter inputs (page 16)
- 4.4 User-defined alarms (page 16)
- 4.5 Product configuration (page 17)
- 4.6 Modules installed (page 17)
- 4.7 Basic settings, CU 372 (page 17)
 - 4.7.1 Display language (page 18)
 - 4.7.2 Units (page 18)
 - 4.7.3 Date and time (page 18)
 - 4.7.4 Password (page 19)
 - 4.7.5 Ethernet (page 19)
 - 4.7.6 GENIbus number (page 19)
 - 4.7.7 Software status (page 19)

12. Status (1)

When the control cabinet is switched on, this status display will appear. No settings can be made in this menu.

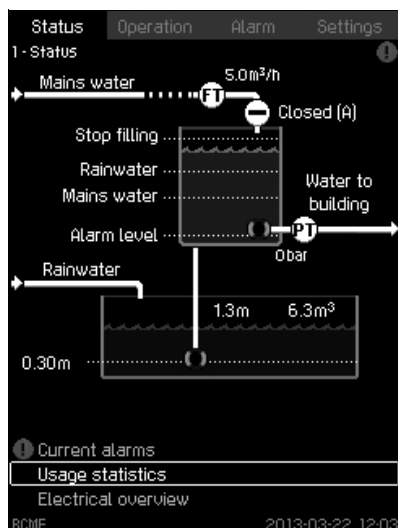


Fig. 8 Status

If the buttons on the control panel have not been touched for 15 minutes, the display will return to this display, and the back light of the panel will be dimmed.

12.1 Electrical overview (1.1)

This display shows an overview of the various inputs and outputs. The submenus in this display are described in the following sections.

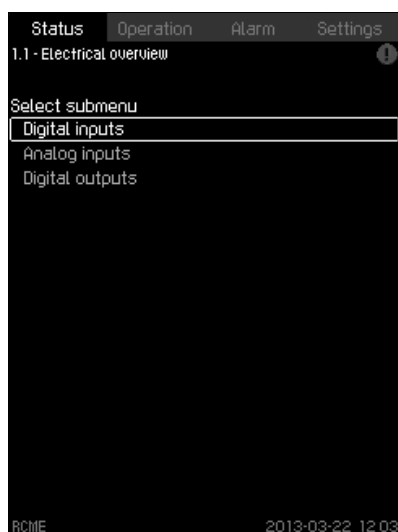


Fig. 9 Electrical overview

12.1.1 Analog inputs (1.1.1)

This display shows an overview of the individual analog inputs and the current measured value of each input.

The factory configuration is shown in fig. 10.

Example

The AI1 analog input on the CU 372 (designated terminal 51) is linked to the input "Level in collecting tank".

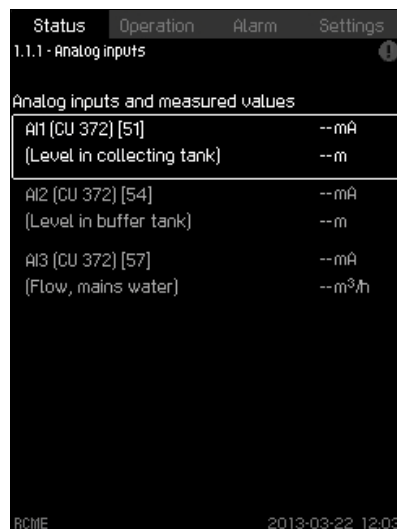


Fig. 10 Analog inputs

12.1.2 Digital inputs (1.1.2)

This display shows the status of the individual digital inputs.

The factory configuration is shown in fig. 11.

Example

The DI1 digital input on the CU 372 (designated terminal 10) is linked to the input "Fault signal from feed pump".

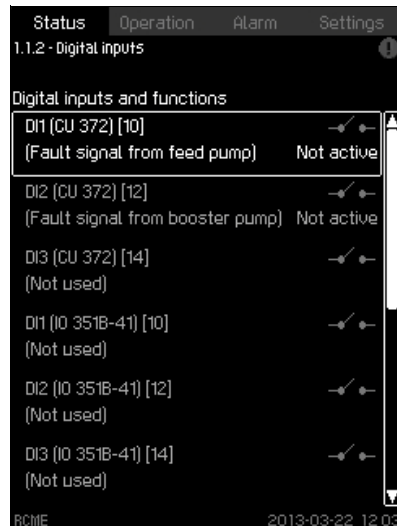


Fig. 11 Digital inputs

Display_1.1.1

Display_1.1.2

12.1.3 Digital outputs (1.1.3)

This display shows the status of the individual digital outputs. The factory configuration is shown in fig. 12.

Example

The DO1 digital output on the CU 372 (designated terminal 71) is linked to the output "Open mains water valve".

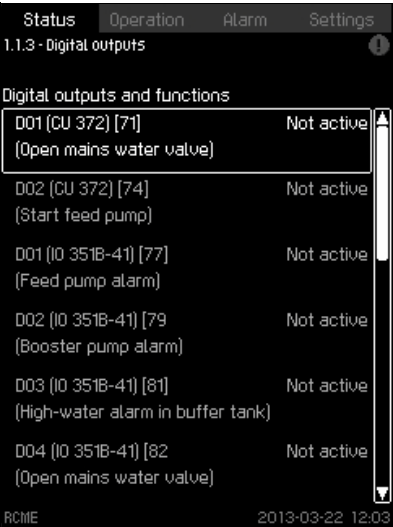


Fig. 12 Digital outputs

12.2 Usage statistics (1.2)

This display shows statistics for the mains water valve and booster pump.

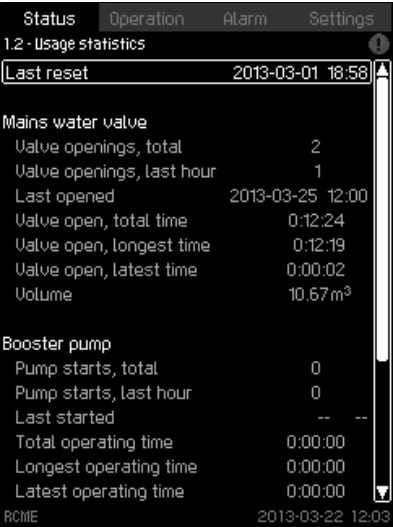


Fig. 13 Usage statistics

13. Operation (2)

In this display, it is possible to change the operating status of the mains water valve, booster pump and the feed pump. Furthermore, the emptying of the buffer tank can be initiated, for example for service purposes. See also section 15.2 *Booster pump settings* (4.2).

Note *In case of an interruption of the power supply, the system will be started with the settings most recently made.*

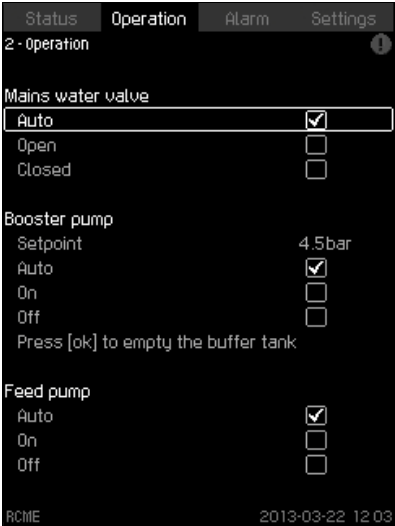


Fig. 14 Operation

14. Alarm (3)

From this display, it is possible to access a list of current alarms and the alarm log.

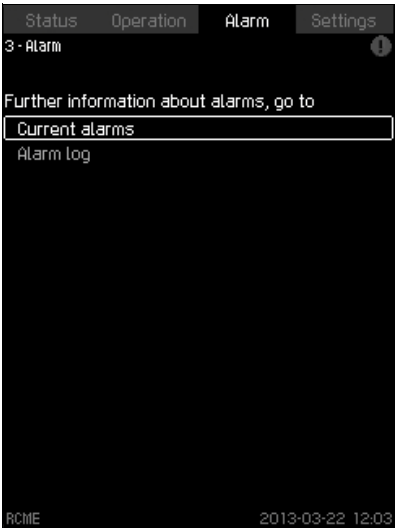


Fig. 15 Alarm

14.1 Current alarms (3.1)



Display_3.1

Fig. 16 Current alarms

This menu shows the following:

- Warnings ⚠ caused by faults that still exist.
- Warnings ⚠ caused by faults that have disappeared, but the warning requires manual resetting.
- Alarms ⓘ caused by faults that still exist.
- Alarms ⓘ caused by faults that have disappeared, but the alarm requires manual resetting.

All warnings and alarms with automatic resetting are automatically removed from the display when the fault has been corrected.

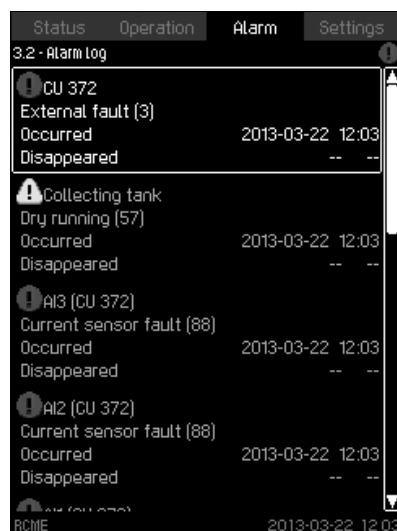
Alarms requiring manual resetting are reset in this display by pressing [OK]. An alarm cannot be reset until the fault has been corrected.

For every warning or alarm, the following is shown:

- Whether it is a warning ⚠ or an alarm ⓘ.
- Where the fault occurred: Feed pump alarm, Booster pump alarm, etc.
- In the case of input-related faults, the input will be shown.
- What the cause of the fault is, and the alarm code in brackets: Water shortage (214), Pressure above maximum pressure (210), etc.
- When the fault occurred: Date and time.
- When the fault disappeared: Date and time. If the fault still exists, date and time are shown as --:--:--.

The latest warning or alarm is shown at the top of the display.

14.2 Alarm log (3.2)



Display_3.2

Fig. 17 Alarm log

This menu shows the following:

- Warnings ⚠ caused by faults that still exist.
- Alarms ⓘ caused by faults that still exist.
- Alarms ⓘ caused by faults that have disappeared, but the alarm requires manual resetting.

All warnings and alarms with automatic resetting are automatically removed from the display when the fault has been corrected.

Alarms requiring manual resetting are reset in this display by pressing [OK]. An alarm cannot be reset until the fault has been corrected.

For every warning or alarm, the following is shown:

- Whether it is a warning ⚠ or an alarm ⓘ.
- Where the fault occurred: Feed pump alarm, Booster pump alarm, etc.
- In the case of input-related faults, the input will be shown.
- What the cause of the fault was, and the alarm code in brackets. This is very important to enable the Grundfos service engineer to identify and correct the fault.
- When the fault occurred: Date and time.
- When the fault disappeared: Date and time. If the fault still exists, date and time are shown as --:--:--.

The latest warning or alarm is shown at the top of the display.

15. Settings (4)

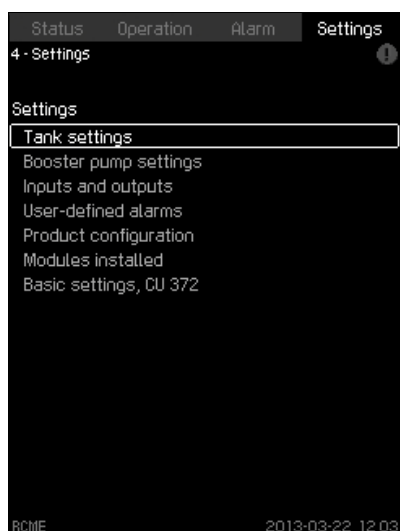


Fig. 18 Settings

In this menu, it is possible to access the following submenus in order to change settings.

Submenus:

- Tank settings
- Booster pump settings
- Inputs and outputs
- User-defined alarms
- Product configuration
- Modules installed
- Basic settings, CU 372.

15.1 Tank settings (4.1)

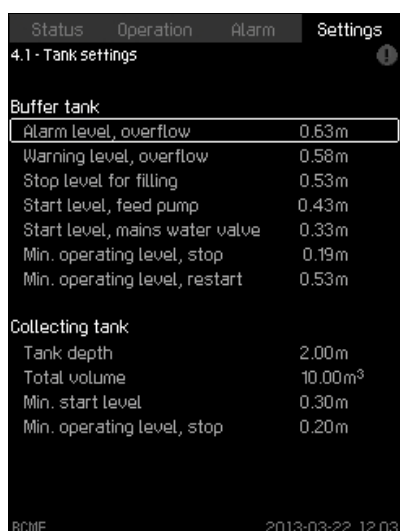


Fig. 19 Tank settings

In this menu, settings can be made for the buffer tank and the collecting tank.

15.1.1 Buffer tank settings

Note *The levels for the buffer tank have been set correctly from the factory and should not be changed.*

- Alarm level, overflow
At this level, an alarm for overflow is activated. The level is factory-set and cannot be changed.
- Warning level, overflow
At this level, a warning for overflow is activated as there is a risk of overflow. The level is factory-set and cannot be changed.
- Stop level for filling
At this level, the filling of the buffer tank must stop. The tank is filled either by the feed pump or via the mains water valve. The level is factory-set and cannot be changed.
- Start level, feed pump
At this level, the feed pump starts supplying rainwater to the buffer tank. The level can be changed.
- Start level, mains water valve
At this level, the mains water valve opens, and mains water is supplied to the buffer tank. The level can be changed.
- Min. operating level, stop
At this level, the booster pump is no longer allowed to run and is stopped. The level is factory-set and cannot be changed.
- Min. operating level, restart
At this level, the booster pump is allowed to run after a dry-running situation. The level can be changed.

Factory settings

Level	Setting
Alarm level, overflow (fixed value)	0.63 m
Warning level, overflow (fixed value)	0.58 m
Stop level for filling (fixed value)	0.53 m
Start level, feed pump	0.43 m
Start level, mains water valve	0.33 m
Min. operating level, stop (fixed value)	0.19 m
Min. operating level, restart	0.43 m

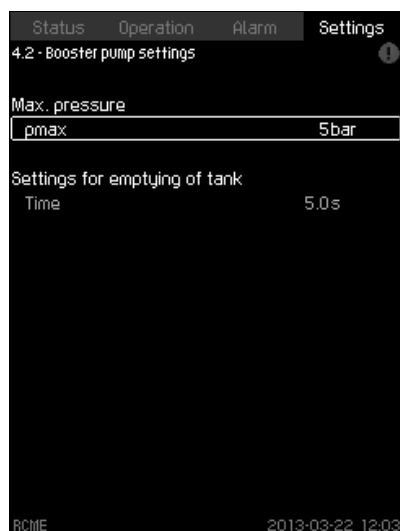
15.1.2 Collecting tank settings

- Tank depth
Enter the depth of the collecting tank.
- Total volume
Enter the total volume of the collecting tank.
- Min. start level
Enter the minimum level at which the feed pump is allowed to start.
- Min. operating level, stop
Enter the minimum operating level at which the feed pump is no longer allowed to run and will be stopped.

Factory settings

Level	Setting
Tank depth	2.00 m
Total volume	10.00 m ³
Min. start level	0.30 m
Min. operating level, stop	0.20 m

15.2 Booster pump settings (4.2)



Display_4.2

Fig. 20 Booster pump settings

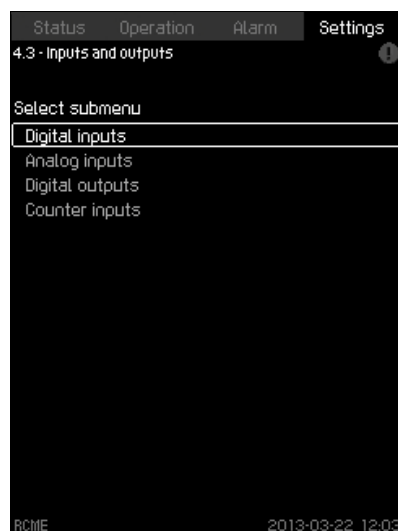
In this menu, the maximum allowed discharge pressure ($p_{\max}$) can be set with 1 bar increments. The chosen setting prevents the setpoint from being set to a value higher than $p_{\max}$.

Under "Settings for emptying of tank", it is possible to set how long the booster pump is allowed to run after "Min. operating level, stop" has been reached. For further information about emptying of the buffer tank, see section [13. Operation \(2\)](#).

Factory setting

$p_{\max}$	5 bar
Time	5 seconds

15.3 Inputs and outputs (4.3)



Display_4.3

Fig. 21 Inputs and outputs

This menu has these submenus from which the inputs and outputs can be configured:

- Digital inputs
- Analog inputs
- Digital outputs
- Counter inputs.

As standard, the CU 372 has a number of inputs and outputs.

As an option, an IO 351B module can be installed to increase the number of inputs and outputs.

Number of possible digital/analog inputs and digital outputs:

Type	Number of inputs and outputs	
	Without IO 351B module	With IO 351B module
Digital inputs	3	12
Analog inputs	3	5
Digital outputs	2	9

15.3.1 Digital inputs (4.3.1)

In this menu, the digital inputs of the CU 372 and IO 351B can be set.

The inputs DI1, DI2 and DI3 of the CU 372 have been factory-configured, as has input DI1 of the IO 351B module, if installed. The remaining eight inputs can be defined by the user.

In the display, all digital inputs are shown so that their physical position in the control cabinet can be identified.

Example

DI1 (IO 351B-41) [10]:

DI1	Digital input 1
IO 351B-41	IO 351B, GENIbus number 41
[10]	Terminal 10

For further information on the connection of the various digital inputs, see the wiring diagram supplied with the control cabinet.

This display shows the three digital inputs of the CU 372 and the nine digital inputs of the IO 351B, if installed.

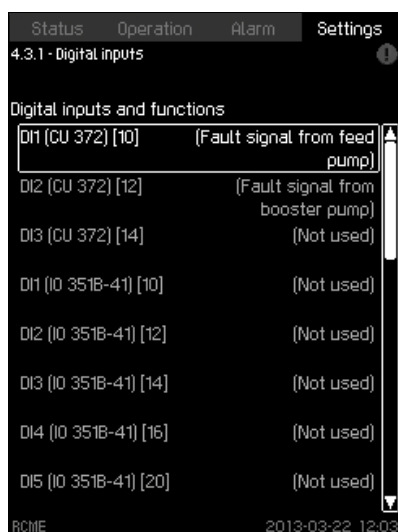


Fig. 22 Digital inputs

Standard configuration:

DI1 (CU 372) [10]	Fault signal from feed pump
DI2 (CU 372) [12]	Fault signal from booster pump
DI3 (CU 372) [14]	Not used
DI1 (IO 351B-41) [10]	Not used
DI2 (IO 351B-41) [12]	Not used
DI3 (IO 351B-41) [14]	Not used
DI4 (IO 351B-41) [16]	Not used
DI5 (IO 351B-41) [20]	Not used
DI6 (IO 351B-41) [24]	Not used
DI7 (IO 351B-41) [42]	Not used
DI8 (IO 351B-41) [44]	Not used
DI9 (IO 351B-41) [46]	Not used

15.3.2 Digital inputs and functions (4.3.1.1)

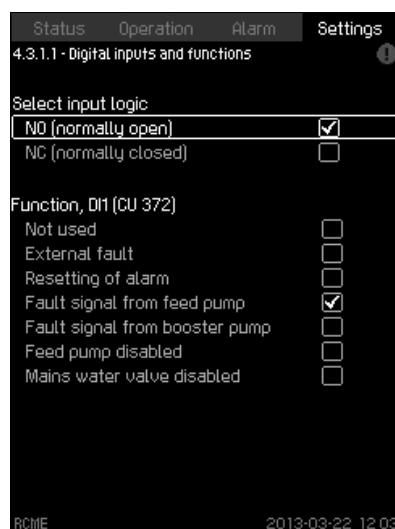


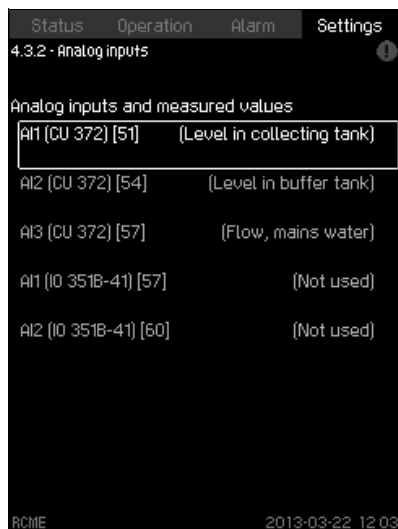
Fig. 23 Digital inputs and functions

In this display a function can be related to the digital inputs.

Setting range

Input logic
NO (normally open)
NC (normally closed)
Functions
Not used
External fault
Resetting of alarm
Fault signal from feed pump
Fault signal from booster pump
Feed pump disabled
Mains water valve disabled

15.3.3 Analog inputs (4.3.2)



Display_4.3.2

Fig. 24 Analog inputs

This display shows the three analog inputs of the CU 372 and two analog inputs of the IO 351B, if installed. Only sensors measuring the following parameters can be connected to the system:

- Level in collecting tank
- Level in buffer tank
- Flow, mains water.

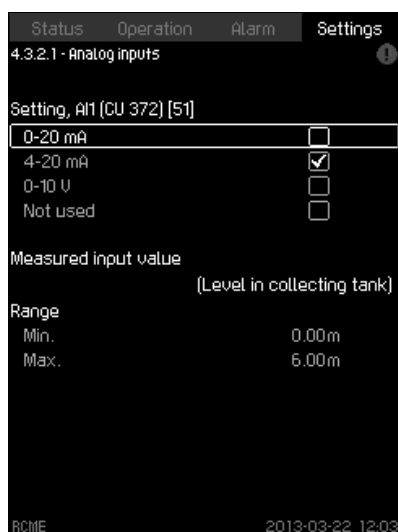
Standard configuration:

AI1 (CU 372) [51]	Level in collecting tank
AI2 (CU 372) [54]	Level in buffer tank
AI3 (CU 372) [57]	Flow, mains water
AI1 (IO 351B-41) [57]	Not used
AI2 (IO 351B-41) [60]	Not used

15.3.4 Analog inputs (4.3.2.1 - 4.3.2.3)

Configure the analog inputs in these displays.

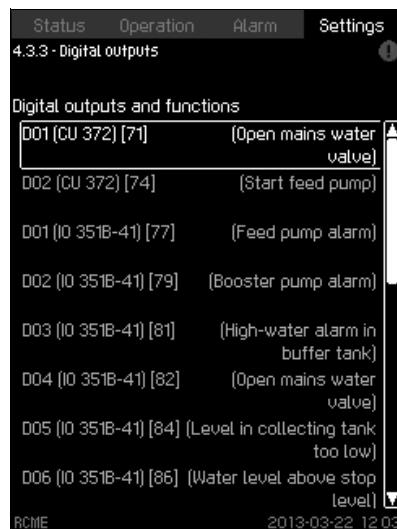
The display below shows an analog input which has been configured for the measurement of the level in the collecting tank.



Display_4.3.2.1

Fig. 25 Analog inputs

15.3.5 Digital outputs (4.3.3)



Display_4.3.3

Fig. 26 Digital outputs

This display shows the two digital outputs of the CU 372 and the seven digital outputs of the IO 351B, if installed.

Standard configuration:

DO1 (CU 372) [71]	Open mains water valve
DO2 (CU 372) [74]	Start feed pump
DO1 (IO 351B-41) [77]	Feed pump alarm
DO2 (IO 351B-41) [79]	Booster pump alarm
DO3 (IO 351B-41) [81]	High-water alarm in buffer tank
DO4 (IO 351B-41) [82]	Open mains water valve
DO5 (IO 351B-41) [84]	Level in collecting tank too low
DO6 (IO 351B-41) [86]	Water level above stop level
DO7 (IO 351B-41) [88]	No function

15.3.6 Counter inputs (4.3.4)

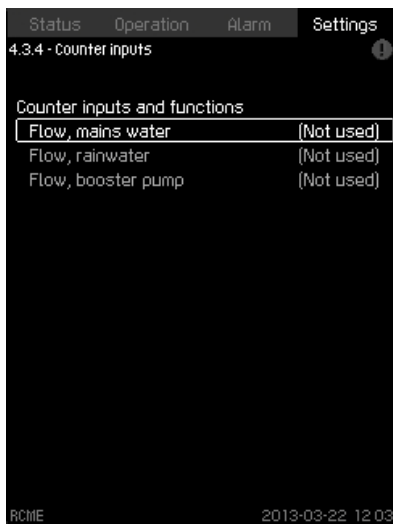


Fig. 27 Counter inputs

This display shows the counter inputs and functions of the CU 372.

Available counter inputs:

- Flow, mains water
- Flow, rainwater
- Flow, booster pump.

15.3.7 Counter inputs (4.3.4.1 - 4.3.4.3)

Configure the counter inputs in these displays.

This display shows an example of the configuration of a digital flowmeter measuring the mains water consumption.

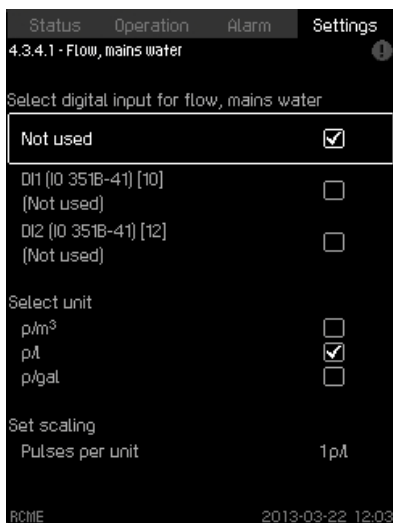


Fig. 28 Flow, mains water

15.4 User-defined alarms (4.4)

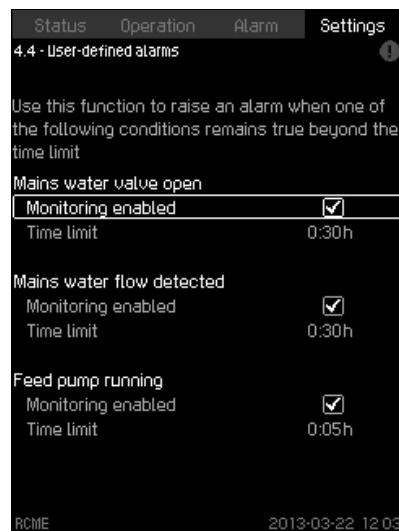


Fig. 29 User-defined alarms

In this menu, it is possible to enable and disable monitoring of the following conditions:

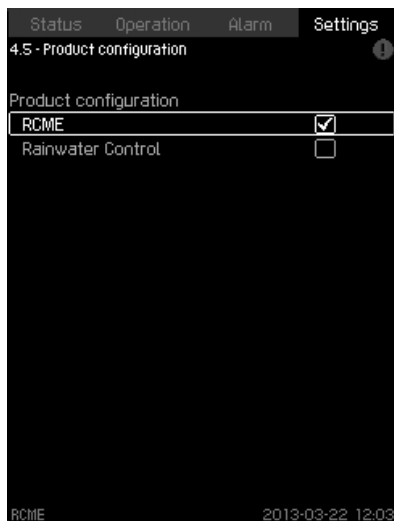
- Mains water valve open
- Mains water flow detected
- Feed pump running.

For each of the monitored conditions, a time limit can be set. The time limit determines the delay between detection of the condition and activation of the alarm. In case of an alarm, the system will stop.

Factory setting

Monitoring enabled.

15.5 Product configuration (4.5)



Display_4.5

Fig. 30 Product configuration

The setting in this display must be made according to the actual system. Under normal conditions, the factory setting does not need to be changed.

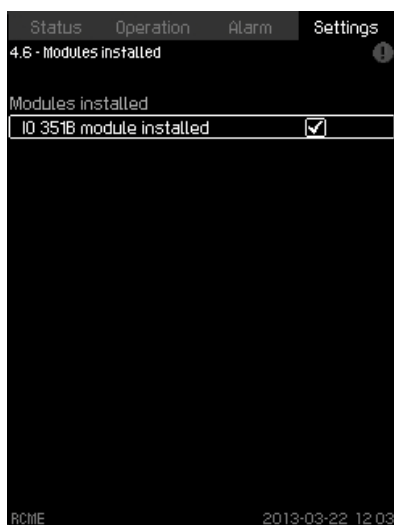
If the system is a complete Grundfos RCME system, select "RCME".

If the system has been specially built for the specific application and is controlled by a Grundfos "Rainwater Control", select "Rainwater Control".

15.6 Modules installed (4.6)

If the tick box in this display is checked, the number of inputs and outputs will be increased if an IO 351B module is installed.

See section [15.3 Inputs and outputs \(4.3\)](#).



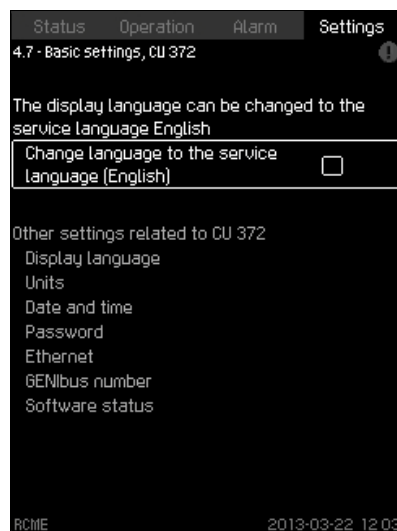
Display_4.6

Fig. 31 Modules installed

15.7 Basic settings, CU 372 (4.7)

In this menu, the basic settings of the CU 372 can be made:

- Activation of service language, English
- Other settings related to CU 372
 - Display language
 - Units
 - Date and time
 - Password for "Operation" and "Settings" menus
 - Ethernet
 - GENIbus number
 - Software status.



Display_4.7

Fig. 32 Basic settings, CU 372

Note

If the service language is selected, the symbol  is shown in the upper right corner of all displays.

15.7.1 Display language (4.7.1)



Fig. 33 Display language

Select the display language of the CU 372.

Possible settings:

- English
- German
- French
- Dutch
- Danish.

If the buttons are not touched for 15 minutes, the display will return to the language previously selected.

Factory setting

The display language is English. It can be changed at start-up.

15.7.2 Units (4.7.2)

Select measuring units for the various parameters shown in the display.

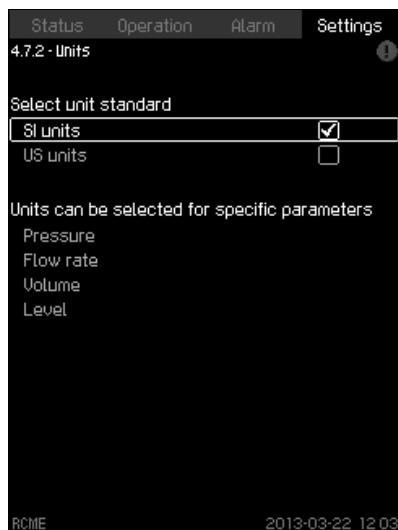


Fig. 34 Units

For the basic setting, choose between SI and imperial units. It is also possible to select other units for the individual parameters.

Setting range

Parameter	Basic setting		Possible units
	SI	Imperial	
Pressure	bar	psi	kPa, MPa, mbar, bar, m, psi
Flow rate	m ³ /h	gpm	m ³ /s, m ³ /h, l/s, gpm, yd ³ /s, yd ³ /min, yd ³ /h
Volume	m ³	gal	l, m ³ , gal, yd ³
Level	m	ft	m, cm, ft, in

Note If units are changed from SI to imperial and vice versa, all individually set parameters are changed to the basic setting in question.

Factory setting

The CU 372 has been factory-set to SI units.

15.7.3 Date and time (4.7.3)

Set the date and time and select how they are to be shown in the display.



Fig. 35 Date and time

The clock has a built-in rechargeable power supply which can supply the clock for up to 20 days if the power supply to the CU 372 is interrupted.

If the clock has been without power for more than 20 days, it must be set again.

Setting range

The date can be set as day, month and year. The time can be set as a 24-hour clock showing hours and minutes.

There are three formats:

Examples of format

2013-03-21 11:49
21-03-2013 11:49
3/21/2013 11:49 am

Factory setting

Local time.

Note If the CU 372 has been without power for more than 20 days since it left the factory, the clock may have returned to the factory setting.

Date and time may have been changed during the setting of the CU 372.

There is no automatic changeover to and from daylight-saving time.

15.7.4 Password (4.7.4)

In this display, it is possible to limit the access to the "Operation" and "Settings" menus by means of a password.



Display_4.7.4

Fig. 36 Password

If the access is limited, it will not be possible to view or set any parameters in the menus.

The password must consist of four digits and may be used for both menus.

Note *If you have forgotten your password, contact Grundfos.*

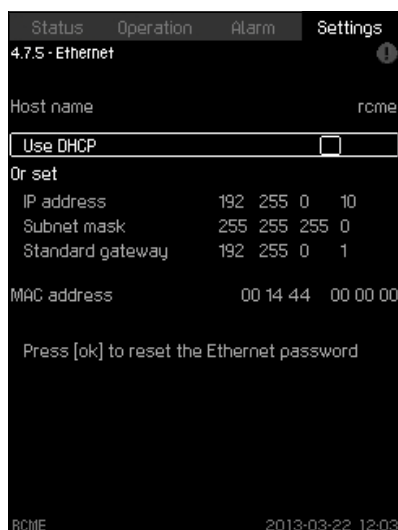
Setting via control panel

1. Select the password to be enabled and press [OK].
2. Select "Enter password" and press [OK].
Now the first digit of the password is flashing.
3. Select digit with [+] and [-] and save with [OK].
Now the second digit of the password is flashing.
4. Repeat points 1 to 3 if another password is to be enabled.

Factory setting

Both passwords are disabled. If a password is to be enabled, the factory setting will be "1234".

15.7.5 Ethernet (4.7.5)



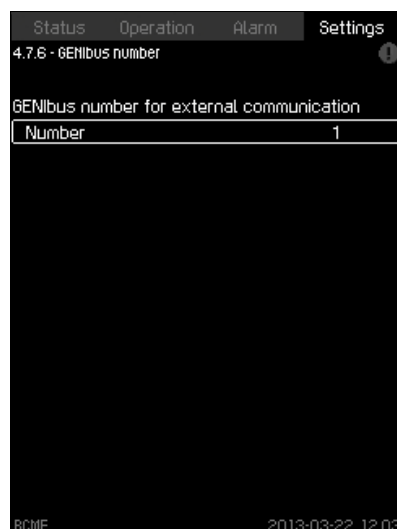
Display_4.7.5

Fig. 37 Ethernet

The CU 372 has an Ethernet connection for communication with a computer, either directly or via the Internet. For further information, see section [16.1 Ethernet](#).

15.7.6 GENIbus number (4.7.6)

In this display, a GENIbus number for external communication can be set.



Display_4.7.6

Fig. 38 GENIbus number

The RCME system can be connected to a network and become part of a superior building management system. For further information, see fig. 40 and section [16.2 GENIbus](#).

The communication is carried out according to the Grundfos bus protocol, GENIbus, and enables connection to a building management system or another external control system.

Operating parameters can be set via the bus signal.

Furthermore, status of important parameters and fault indications can be read from the CU 372.

For further information, contact Grundfos.

Setting range

The number can be set between 1 and 64.

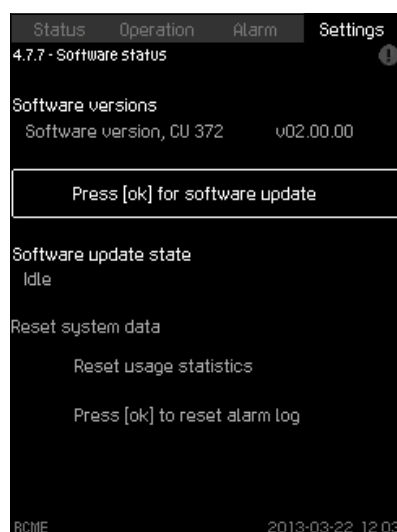
Setting via control panel

1. Select "GENIbus number" with [+] and [-] and press [OK].
2. Select number with [+] and [-] and save with [OK].

Factory setting

41.

15.7.7 Software status (4.7.7)



Display_4.7.7

Fig. 39 Software status

This display shows the version of the software installed in the CU 372.

In this display, it is possible to reset usage statistics. See section [12.2 Usage statistics \(1.2\)](#). The alarm log can also be reset in this display.

As it is a status display, no settings can be made.

16. Data communication

The CU 372 incorporates hardware that enables communication with external units, such as a computer, via an external GENIbus or Ethernet connection.

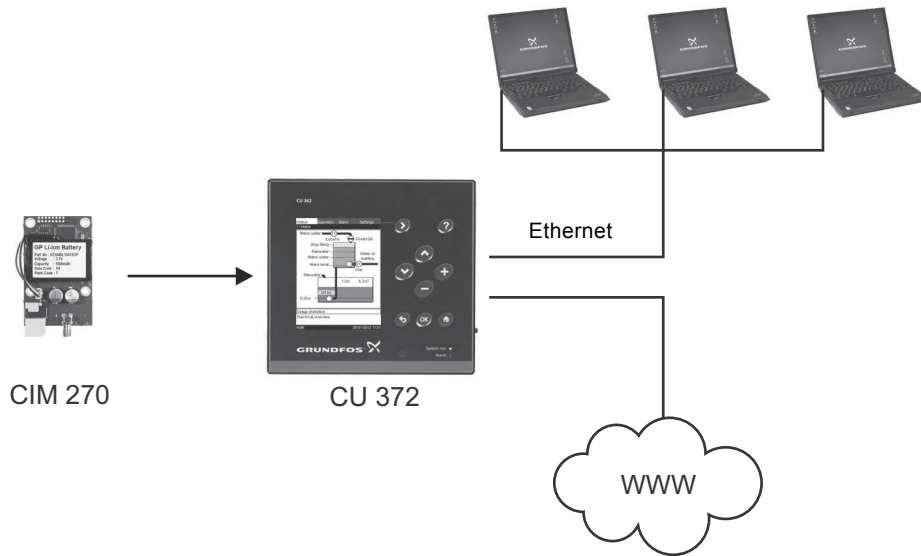


Fig. 40 Data communication via external GENIbus and Ethernet connections

16.1 Ethernet

Ethernet is the most widely used standard for local area networks (LAN). The standardisation of this technology has created some of the easiest and cheapest ways of creating communication between electrical units, for instance between computers or between computers and control units.

The web server of the CU 372 makes it possible to connect a computer to the CU 372 via an Ethernet connection. The user interface can thus be exported from the CU 372 to a computer so that the CU 372 and consequently the RCME system can be remotely monitored and controlled.

Note

We recommend that you protect the connection to the CU 372 in consultation with the system administrator according to your safety requirements.

In order to use the web server, you must know the IP address of the CU 372. All network units must have a unique IP address to be able to communicate with each other.

It is possible to change the factory-set IP address by configuration or by activating a DHCP (Dynamic Host Configuration Protocol), either directly in the CU 372 or via the web server. See the example in fig. 41.



Fig. 41 Example of Ethernet setting

Display_4.7.5

Dynamic assignment of an IP address for the CU 372 requires a DHCP server in the network. The DHCP server assigns a number of IP addresses to the electrical units and makes sure that two units do not receive the same IP address.

Use a standard Internet browser when connecting to the web server of the CU 372.

If you want to use the factory-set IP address, no changes are required in the display. Open the Internet browser, and enter the IP address of the CU 372.

In order to use dynamic assignment, the function must be enabled. Click [OK] when the "Use DHCP" line is highlighted. A tick shows that the function has been enabled. Open the Internet browser, and enter the CU 372 "Host name" instead of the IP address (the host name is stated at the top of display 4.7.5). The Internet browser will now try to establish connection to the CU 372.

DHCP requires a host name.

The host name can only be changed by means of a GSC file (configuration file) or via a web server. See the Change of network setting section, page 21.

Note

This is the first display that appears when connecting to the CU 372.

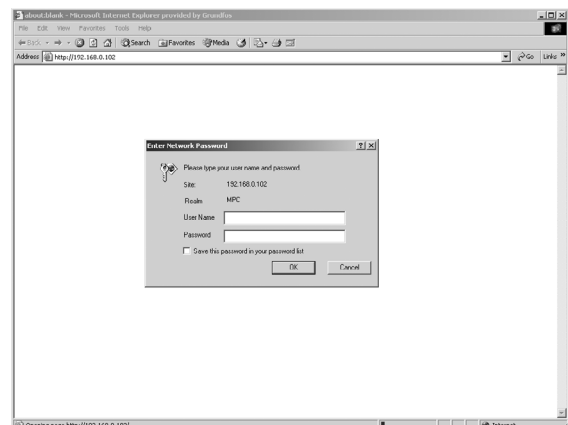


Fig. 42 Connecting to the CU 372

TM05 6926 0313

TM03 2048 3505

Factory setting

User name: admin

Password: admin

When user name and password have been entered, a Java Runtime Environment application starts up in the CU 372, provided that it has been installed on the computer in question. If this is not the case, but the computer is connected to the Internet, then use the link on the screen to download and install the Java Runtime Environment application.

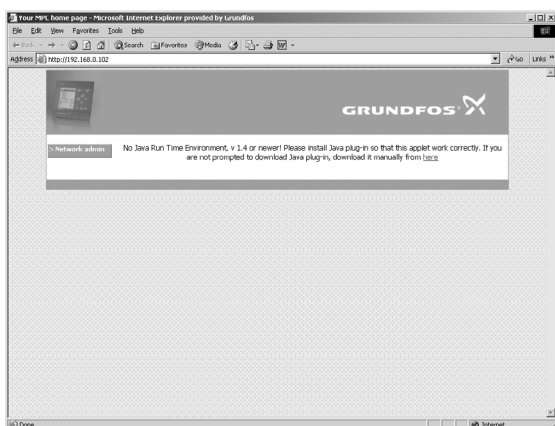


Fig. 43 Display with link to Java Runtime Environment

The Java Runtime Environment application transfers the CU 372 user interface (including display and control panel) to the computer screen. It is now possible to monitor and control the CU 372 from the computer.

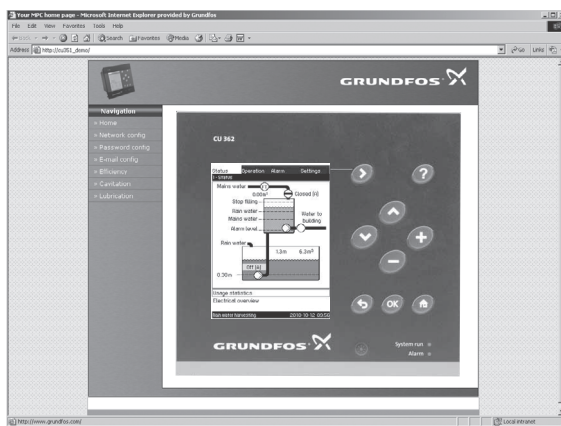


Fig. 44 Network setting

Change of network setting

When the connection to the web server of the CU 372 has been established, it is possible to change the network setting.

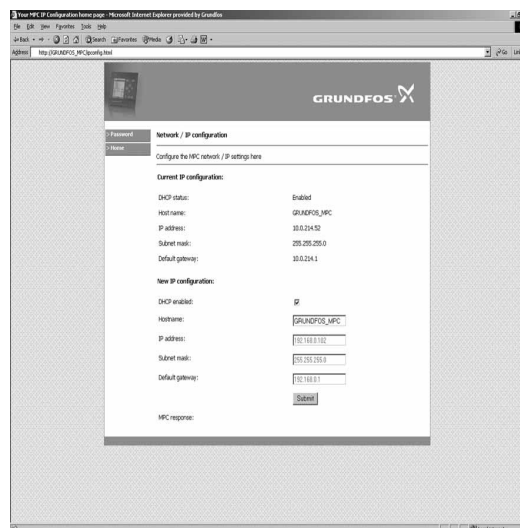


Fig. 45 Change of network setting

1. Click the "Network admin" icon.
2. Enter the changes.
3. Click [Submit] to enable the changes.

Change of password

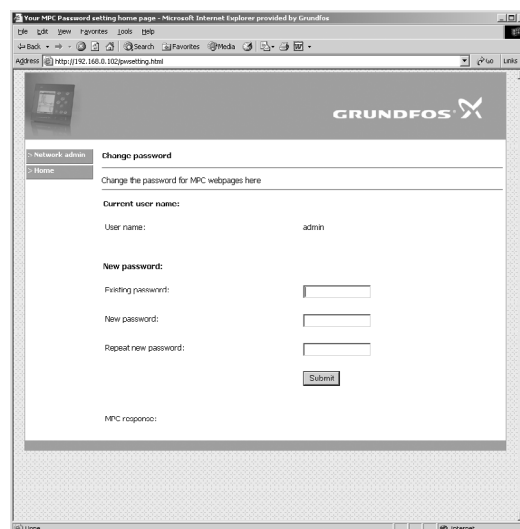


Fig. 46 Change of password

1. Click the "Change password" icon.
2. Enter the new password.
3. Click [Submit] to enable the new password.

16.2 GENIbus

When a GENIbus module is installed, it is possible to connect a CU 372 to an external network. The connection can be established via a GENIbus-based network or a network based on another protocol via a gateway. See fig. 40. For further information, contact Grundfos.

TM03 2050 3505

TM03 2049 3505

TM05 6960 0313

TM03 2051 3505

17. Maintenance



Warning

Before starting any work on pumps, terminal boxes or control cabinet, the power supply must have been switched off for at least five minutes, and it must be ensured ensure that it cannot be accidentally switched on.

17.1 CU 372

The CU 372 is maintenance-free. It must be kept clean and dry. Protect it from direct sunlight. The temperature of the CU 372 must not exceed the maximum ambient temperature.

See section [19.2 Temperatures](#).

17.2 Regular inspections

The RCME system should be inspected at regular intervals. This must be carried out when the system has been shut down.

Perform the following:

- Check that there are no leaks in the internal pipework, the pump and the buffer tank.
- Check the precharge pressure of the pressure vessel and adjust, if necessary. See section [17.3 Integrated pressure vessel](#).
- Check that the booster pump operates quietly and smoothly without vibration or excessive noise.

Should any faults be found, see section [21. Fault finding](#).

After inspection, ensure that the isolating valves are fully open, and that the power supply to the RCME system is switched on again.

17.3 Integrated pressure vessel

Set the precharge pressure of the vessel to 70 % of the setpoint selected on the booster pump control panel (booster pump discharge pressure).

Note

Measure the precharge pressure while the system is pressureless.

If the precharge pressure is not set correctly, this may result in incorrect operation of the RCME unit.

Check and adjust the precharge pressure of the vessel before the booster pump is filled with water. If the system has already been filled and the pump has been running, proceed as follows:

1. Switch off the power supply to the RCME system.
2. Release the water pressure by opening a tap in the discharge pipe of the system. Leave the tap open while checking and adjusting the precharge pressure.
3. The precharge pressure can now be checked using a car tyre pressure gauge. The air valve is located under the small lid on top of the pressure vessel.
For small pressure adjustments a foot pump can be used. If large amounts of gas are required, use dry nitrogen to prevent corrosion within the vessel.

Setting example:

If the setpoint has been set to 4.5 bar, the precharge pressure must be $4.5 \times 0.7 = 3.15$ bar.

18. Taking the RCME system out of operation

Take the RCME system out of operation by switching the main switch off.



Warning

The conductors to the main switch are still live. Lock the main switch to ensure that the power supply cannot be accidentally switched on.

19. Technical data

19.1 Pressure

Maximum operating pressure: 5 bar.

19.2 Temperatures

Liquid temperature: 3-40 °C.

Maximum ambient temperature: 40 °C.

19.3 Relative air humidity

Maximum 95 %.

19.4 Sound pressure level

65 dB(A).

19.5 Enclosure class

IP54.

19.6 Volume of buffer tank

Rated volume: 160 litres.

Effective water volume: 108 litres.

Minimum booster pump operating volume: 41 litres.

19.7 Weight of RCME unit

Weight with filled buffer tank: 200 kg.

The stated weight includes base plate, booster pump and buffer tank.

19.8 Dimensions

See fig. [3](#) on page [25](#).

20. Electrical data

20.1 Supply voltage

3 x 400 VAC - 10 %/+ 10 %, 50 Hz, PE.

21. Fault finding



Warning

Before starting fault finding, switch off the power supply. Make sure that the power supply cannot be accidentally switched on.

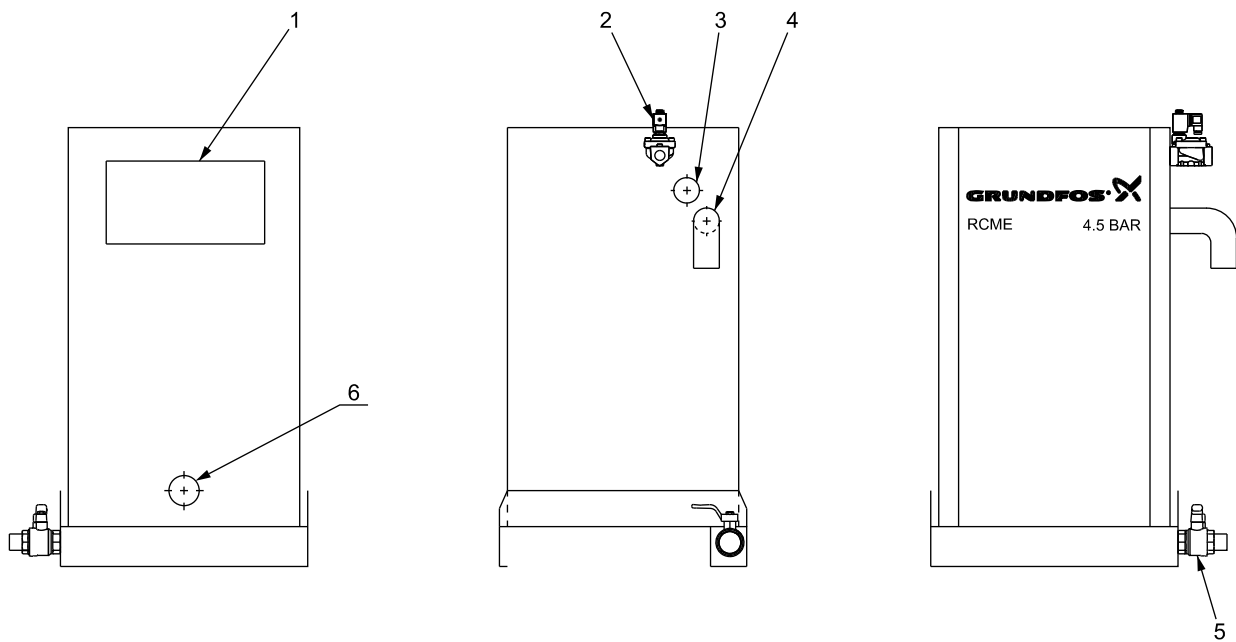
Fault	Cause	Remedy
1. No rainwater is supplied to the buffer tank even though the collecting tank contains rainwater.	a) The feed pump is malfunctioning.	See the installation and operating instructions for the feed pump.
	b) The feed pump has not been set to "Auto".	Set the feed pump to "Auto". See section 13. Operation (2) .
	c) The feed pump discharge pipe is blocked or cracked.	Remove blockage, or replace the pipe.
	d) Level sensor in collecting tank malfunctioning.	Check input from collecting tank level sensor on the status display. See section 12. Status (1) . Check sensor connections and sensor operation. Replace the sensor, if necessary.
	e) Level sensor in buffer tank malfunctioning.	Check input from buffer tank level sensor on the status display. See section 12. Status (1) . Check sensor connections and sensor operation. Replace the sensor, if necessary.
	f) Overload trip activated.	Check reason for overload.
2. The booster pump is running, but the discharge pressure is low.	a) Air in the booster pump.	Take the system out of operation by switching the main switch off. Slacken the vent screw on the back of the pump sleeve and wait until airless water comes out. Retighten the screw.
3. No rainwater is available, and mains water is not being supplied to the buffer tank.	a) The mains water valve has not been set to "Auto".	Set the mains water valve to "Auto". See section 13. Operation (2) .
	b) The mains water valve is malfunctioning.	Check whether the valve is blocked by impurities, and clean it, if necessary. Replace the valve, if necessary.
	c) User-defined alarm is activated.	Reset the alarm. See sections 15.4 User-defined alarms (4.4) and 14.1 Current alarms (3.1) .
4. The booster pump is unable to reach the setpoint.	a) The consumption is too high.	Reduce the consumption, if possible.
	b) Pipe fracture or leakage in the system.	Check the system, and repair, if necessary.
5. No connection to Ethernet.	a) Wrong host name.	Check the host name. See section 16.1 Ethernet .

22. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.



TM05 9761 4413

Fig. 1 Connections on RCME pressure boosting unit

1	GB	Opening for overflow purpose in order to comply with EN 1717
	DK	Åbning til overløb for at kunne overholde EN 1717
	DE	Rechteckiger Ausschnitt als Überlauf gemäß EN 1717.
	FR	Ouverture de trop-plein conformément à la norme EN 1717
	NL	Afschakeling voor overstroomdoeleinden om te voldoen aan EN1717
2	GB	Mains water valve
	DK	Ventil, vandværksvand
	DE	Leitungswasserzulaufventil
	FR	Vanne réseau
3	GB	Rainwater inlet
	DK	Tilgang for regnvand
	DE	Regenwasserzulauf
	FR	Entrée d'eau de pluie
4	GB	Overflow
	DK	Overløb
	DE	Überlauf
	FR	Trop plein
6	GB	Mains water valve
	DK	Ventil, vandværksvand

5	GB	Outlet connection
	DK	Afgangstilslutning
	DE	Abgang
	FR	Raccord de sortie
6	NL	Uitlaataansluiting
	GB	Connection for additional buffer tank
	DK	Tilslutning for ekstra buffertank
	DE	Anschluss für weiteren Speichertank.
5	FR	Raccordement du réservoir tampon additionnel.
	NL	Aansluiting voor extra buffertank.

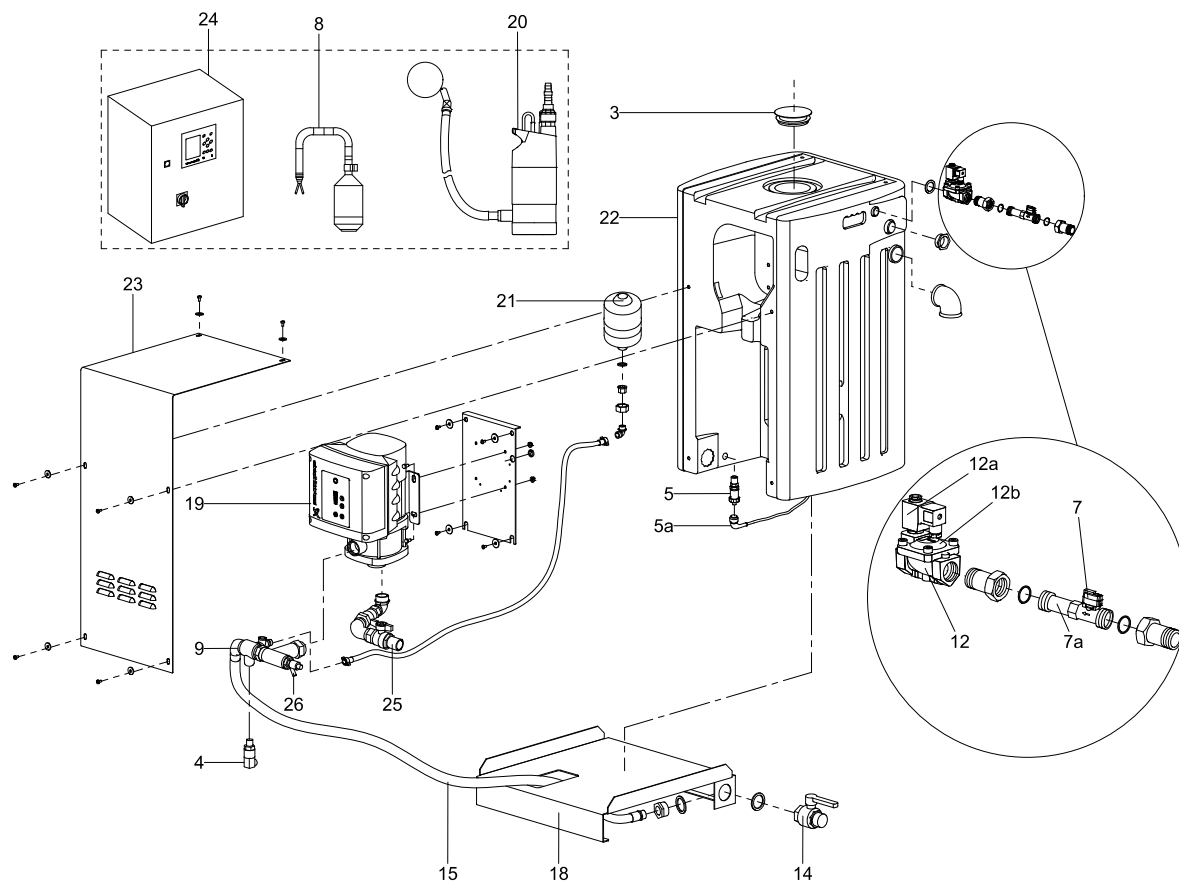


Fig. 2 Exploded view of main components

TM05 9722 4413

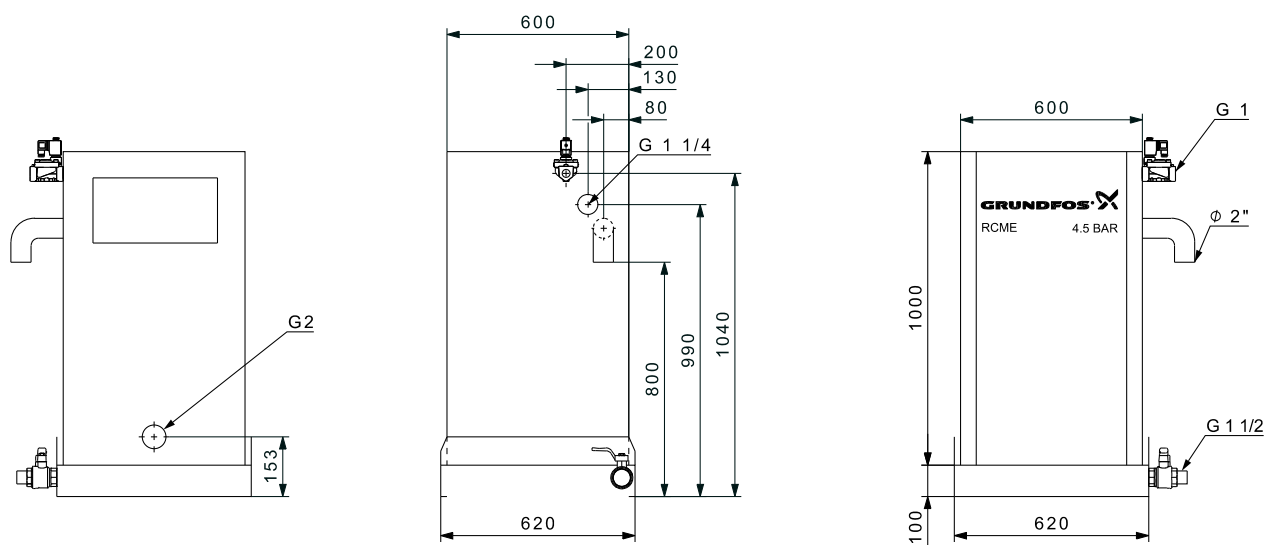


Fig. 3 Dimensional sketch

Dimensions are in mm.

TM04 9195 3913

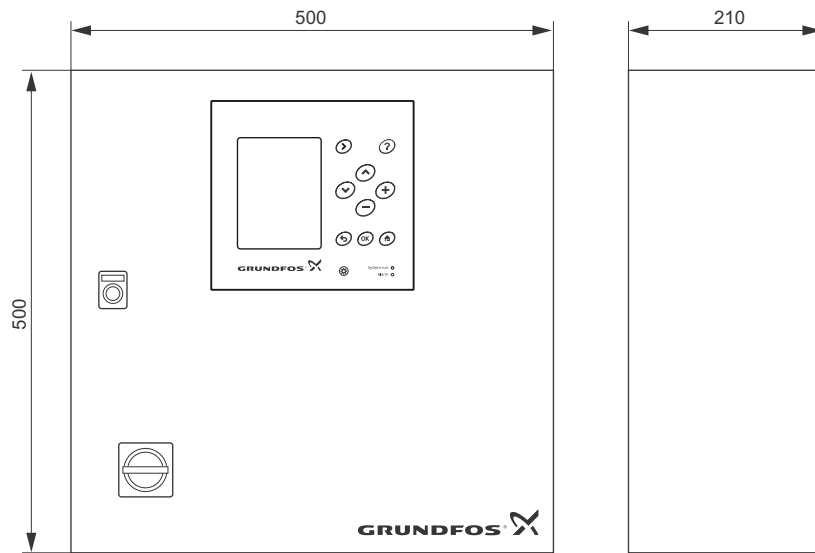


Fig. 4 Dimensional sketch of control cabinet with IO 351B module installed inside

Dimensions are in mm.

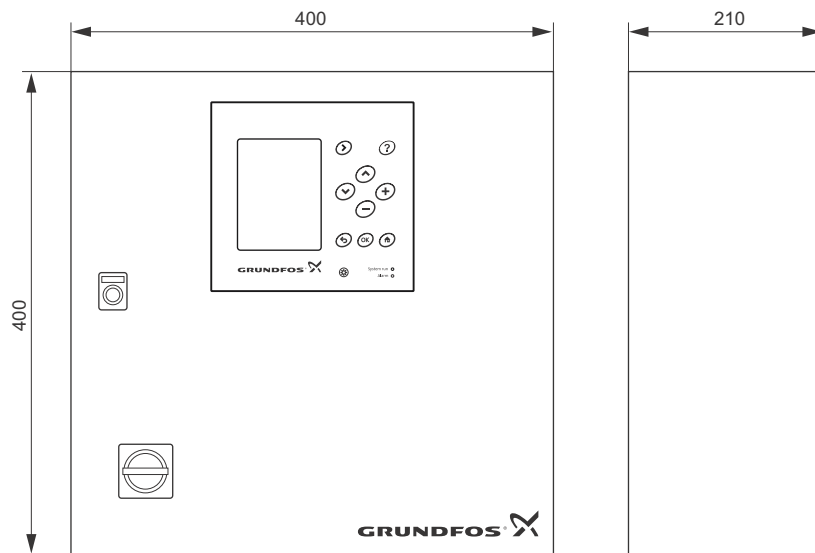


Fig. 5 Dimensional sketch of control cabinet with optional IO 351B module installed inside

TM05 6959 0313

TM05 9723 4413

Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the product RCME, to which this declaration relates, is in conformity with these Council directives on the approximation of the laws of the EC member states:

- Machinery Directive (2006/42/EC).
Standard used: EN809:2008.
 - Low Voltage Directive (2006/95/EC).
Standards used: EN 60204-1:2006 + A1:2009.
 - EMC Directive (2004/108/EC).
Standards used: EN 61000-6-2:2005 and EN 61000-6-3:2007.
- This product is manufactured according to EN 1717 with a type AB air gap and an unrestricted, non-circular opening for overflow.

This EC declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 98449415 0114).

DE: EG-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt RCME, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmt:

- Maschinenrichtlinie (2006/42/EG).
Norm, die verwendet wurde: EN809:2008.
- Niederspannungsrichtlinie (2006/95/EG).
Normen, die verwendet wurden: EN 60204-1:2006 + A1:2009.
- EMV-Richtlinie (2004/108/EG).
Normen, die verwendet wurden: EN 61000-6-2:2005 und EN 61000-6-3:2007.

Das Produkt wurde in Übereinstimmung mit der EN 1717 hergestellt. Es verfügt über eine nicht kreisförmige Öffnung vom Typ AB als Überlauf, der als freier Auslauf ausgeführt ist.

Diese EG-Konformitätserklärung gilt nur, wenn sie in Verbindung mit der Grundfos Montage- und Betriebsanleitung (Veröffentlichungsnummer 98449415 0114) veröffentlicht wird.

NL: EC overeenkomstigheidsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat het product RCME waarop deze verklaring betrekking heeft, in overeenstemming is met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG lidstaten betreffende:

- Machine Richtlijn (2006/42/EC).
Gebruikte norm: EN809:2008.
- Laagspannings Richtlijn (2006/95/EC).
Gebruikte normen: EN 60204-1:2006 + A1:2009.
- EMC Richtlijn (2004/108/EC).
Gebruikte normen: EN 61000-6-2:2005 en EN 61000-6-3:2007.

Dit product wordt vervaardigd conform de EN 1717 met een type AB luchtspleet en een onbepaalde, niet-cirkelvormige opening voor overloop.

Deze EC overeenkomstigheidsverklaring is alleen geldig wanneer deze gepubliceerd is als onderdeel van de Grundfos installatie- en bedieningsinstructies (publicatienummer 98449415 0114).

DK: EF-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produktet RCME som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

- Maskindirektivet (2006/42/EF).
Anvendt standard: EN809:2008.
 - Lavspændingsdirektivet (2006/95/EF).
Anvendte standarder: EN 60204-1:2006 + A1:2009.
 - EMC-direktivet (2004/108/EF).
Anvendte standarder: EN 61000-6-2:2005 og EN 61000-6-3:2007.
- Dette produkt er fremstillet i henhold til EN 1717 med en type AB luftafstand og en uhindret, ikke-rund åbning til overløb.

Denne EF-overensstemmelseserklæring er kun gyldig når den publiceres som en del af Grundfos-sikkerhedsanvisningerne (publikationsnummer 98449415 0114).

FR: Déclaration de conformité CE

Nous, Grundfos, déclarons sous notre seule responsabilité, que le produit RCME, auquel se réfère cette déclaration, est conforme aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous:

- Directive Machines (2006/42/CE).
Norme utilisée: EN809:2008.
- Directive Basse Tension (2006/95/CE).
Normes utilisées: EN 60204-1:2006 + A1:2009.
- Directive Compatibilité Electromagnétique CEM (2004/108/CE).
Normes utilisées: EN 61000-6-2:2005 et EN 61000-6-3:2007.

Ce produit est conforme à la norme EN 1717, type AB, surverse avec trop-plein non circulaire.

Cette déclaration de conformité CE est uniquement valide lors de sa publication dans les consignes de sécurité Grundfos (numéro de publication 98449415 0114).

Bjerringbro, 2nd April 2013



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